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E-worksheet assisted by Liveworksheets: Innovation in learning media for chemical bonds based on ethnoscience to improve critical thinking skills of high school students

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ABSTRACT

The abstract nature of chemical bonding often leads to limited conceptual understanding and weak critical-thinking skills, underscoring the need for innovative, contextually relevant learning media. Therefore, this study integrates an ethnoscience approach with Liveworksheets-assisted interactive tasks to support contextual learning and enhance students' higher-order thinking in chemistry. This study aimed to develop an ethnoscience-based e-worksheet supported by Liveworksheets and to examine its effectiveness in enhancing high school students' critical thinking skills. A Research and Development (R and D) method using the 4D model was implemented, encompassing expert validation, practicality assessment through student response questionnaires, and effectiveness testing using a one-group pretest–posttest design involving 36 eleventh-grade students at a public senior high school. The developed e-worksheet was found to be highly valid and very practical, achieving a practicality score of 96.5%. Effectiveness results indicated a significant improvement in students' critical thinking skills, with an average N-gain of 0.73 in the high category across interpretation, analysis, evaluation, explanation, and inference indicators. These findings demonstrate that integrating ethnoscience and Liveworksheets through e-worksheet promotes meaningful chemistry learning and the development of critical thinking. The findings imply that ethnoscience-based interactive multimedia is an effective instructional resource for supporting contextual and higher-order thinking-oriented chemistry learning.

E-worksheet berbantuan Liveworksheets: Inovasi media pembelajaran ikatan kimia berbasis etnosains untuk meningkatkan kemampuan berpikir kritis siswa sekolah menengah atas

Kata Kunci:

ikatan kimia, berpikir kritis, e-worksheet, etnosains, Liveworksheets

ABSTRAK

Sifat abstrak dari ikatan kimia seringkali menyebabkan keterbatasan pemahaman konseptual dan lemahnya keterampilan berpikir kritis, sehingga menegaskan perlunya media pembelajaran yang inovatif dan kontekstual. Oleh karena itu, penelitian ini mengintegrasikan pendekatan etnosains dengan tugas interaktif berbantuan Liveworksheets untuk mendukung pembelajaran kontekstual dan meningkatkan kemampuan berpikir tingkat tinggi siswa dalam kimia. Penelitian ini bertujuan untuk mengembangkan e-worksheet berbasis etnosains yang didukung oleh Liveworksheets serta menguji keefektifannya dalam meningkatkan keterampilan berpikir kritis siswa SMA. Metode yang digunakan adalah Research and Development (R&D) dengan

model 4D, yang meliputi validasi ahli, uji kepraktisan melalui angket respons siswa, serta uji keefektifan menggunakan desain one-group pretest–posttest yang melibatkan 36 siswa kelas XI di salah satu SMA negeri. E-worksheet yang dikembangkan dinyatakan sangat valid dan sangat praktis, dengan skor kepraktisan sebesar 96,5%. Hasil uji keefektifan menunjukkan adanya peningkatan signifikan pada keterampilan berpikir kritis siswa, dengan rata-rata N-gain sebesar 0,73 yang berada pada kategori tinggi pada indikator interpretasi, analisis, evaluasi, penjelasan, dan inferensi. Temuan ini menunjukkan bahwa integrasi etnosains dan Liveworksheets melalui e-worksheet mampu mendorong pembelajaran kimia yang bermakna serta pengembangan kemampuan berpikir kritis. Hasil penelitian ini mengimplikasikan bahwa multimedia interaktif berbasis etnosains merupakan sumber pembelajaran yang efektif untuk mendukung pembelajaran kimia yang kontekstual dan berorientasi pada keterampilan berpikir tingkat tinggi.

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Contribution to the literature

This research contributes to:

- It proposes an integrated instructional design that combines ethnoscience-based cultural contexts with interactive digital multimedia (e-worksheet) for learning chemical bonding, offering a novel approach that connects local cultural values such as *gotong royong* with abstract chemical concepts.
- The study provides empirical evidence, based on indicators, that ethnoscience-integrated interactive media can systematically enhance students' critical thinking skills in chemical bonding, particularly in understanding covalent and metallic bonds.
- The developed e-worksheet demonstrates how culturally responsive digital learning media can operationalize the principles of the Merdeka Curriculum by promoting contextual, student-centered, and higher-order thinking-oriented chemistry learning.

1. INTRODUCTION

Education is a fundamental process in shaping the knowledge, character, and competencies of individuals who contribute to the advancement of Indonesian society [1]. Through formal and informal channels, education develops important soft skills such as communication, collaboration, leadership, and adaptability, as well as hard skills, including technical, analytical, and language abilities [2]. In chemistry education, these skills are increasingly important for addressing the dynamic challenges of 21st-century learning, as students must interpret abstract chemical concepts and apply them to problem-solving situations [3]. Therefore, improving the quality of education has become a crucial priority in response to rapid technological advancement, digitalization, and globalization.

Amid rapid technological advancement, digitalization, and globalization, improving educational quality has become imperative. In response, the United Nations established the Sustainable Development Goals (SDGs), particularly SDG 4 (Quality Education), which promotes inclusive, equitable, and lifelong learning opportunities [4]. This vision aligns with Indonesia's Independent Curriculum (Kurikulum Merdeka), which emphasizes flexible, contextual, and student-centered learning to strengthen lifelong competencies such as critical thinking, creativity, collaboration, and technological literacy [5]. In chemistry education, these policy directions require innovative instructional approaches

that help students understand abstract chemical concepts through contextual and inquiry-based learning [6]–[8]. Such competencies are particularly important in chemistry, where students must interpret abstract concepts to explain molecular interactions and chemical phenomena.

Chemical bonding is a foundational topic in chemistry because it underlies molecular structure, material properties, and chemical reactions. However, concepts such as covalent and metallic bonding are abstract and involve unobservable microscopic representations, including electron interactions and intermolecular forces [9]. This abstract nature makes it challenging for students to visualize and internalize key concepts effectively. Consequently, students often rely on memorization rather than conceptual understanding. Questionnaire data from eleventh-grade students in Sidoarjo Regency revealed that 70.6% experienced difficulty understanding chemical bonding, particularly distinguishing covalent and metallic bonds and relating them to everyday phenomena, resulting in suboptimal learning outcomes.

The difficulty students experience in understanding chemical bonding is influenced not only by the material's abstract nature but also by instructional factors. In many classrooms, chemical bonding is often taught through teacher-centered explanations and symbolic representations, with limited opportunities for students to visualize submicroscopic processes or relate them to real-life contexts. Such conventional approaches may lead students to rely on memorization rather than conceptual understanding, which in turn limits the development of critical thinking skills. Therefore, innovative instructional approaches are needed to support meaningful learning of abstract chemistry concepts [9]. One promising approach is the integration of ethnoscience and interactive digital media. Ethnoscience connects scientific concepts with local cultural knowledge, enabling students to understand abstract ideas through familiar cultural experiences. At the same time, interactive multimedia can provide visual and participatory learning experiences that help students actively explore complex chemical concepts [10]. Previous studies have shown that ethnoscience-based instruction can enhance students' engagement, contextual understanding, and critical thinking in science learning. Similarly, research on interactive multimedia demonstrates its effectiveness in improving students' comprehension of abstract chemistry concepts and supporting higher-order thinking.

Previous studies have examined ethnoscience and multimedia learning separately. Ethnoscience-based instruction has been shown to enhance critical thinking and contextual understanding in chemistry learning [11]–[15], while multimedia tools enhance understanding of abstract concepts and support higher-order thinking [16]. However, most studies treat these approaches independently, rarely integrating cultural contextualization into digital multimedia learning. Empirical studies integrating ethnoscience into multimedia learning through systematic instructional design, validated learning instruments, and digital platforms for classroom implementation remain limited, particularly within the Independent Curriculum framework and the SDG 4 goals [17]. To address this gap, the present study integrates ethnoscience contexts with interactive multimedia by developing an electronic learning activity and practice document (e-worksheet) [10], [18]. The ethnoscience approach connects chemical bonding concepts with familiar cultural practices, such as the principle of *gotong royong* (cooperation) to illustrate electron sharing in covalent bonding, and traditional Javanese gamelan made of bronze alloys to represent metallic bonding through the "sea of electrons" model. By embedding these cultural analogies within interactive digital activities, the developed e-worksheet aims to transform abstract chemical bonding concepts into contextual, meaningful learning experiences while also promoting students' critical thinking skills.

Therefore, a significant research gap exists in the integrated development and evaluation of ethnoscience-based interactive multimedia e-worksheet for chemical bonding. This study aims to develop and evaluate a Liveworksheets-assisted ethnoscience-based e-worksheet to enhance senior high school students' critical thinking skills in chemical bonding. This study contributes to chemistry education by demonstrating that integrating cultural context and interactive digital learning can address challenges in understanding abstract chemical concepts.

2. METHOD

This study employed a Research and Development (R&D) approach to design and evaluate ethnoscience-based interactive multimedia for instruction on chemical bonding. The R&D method was selected because it not only produces an educational product but also systematically validates its quality, practicality, and effectiveness for eleventh-grade chemistry learning through iterative analysis and evaluation [19], [20].

This development follows the 4D model by Thiagarajan, Semmel, and Semmel, including Define, Design, Develop, and Dissemination and also provides a structured framework for producing valid and effective learning media through gradual refinement [21], [22]. However, this study was limited to three stages: Define, Design, and Develop, conducted over four months to ensure focused validation and practicality testing before broader implementation.

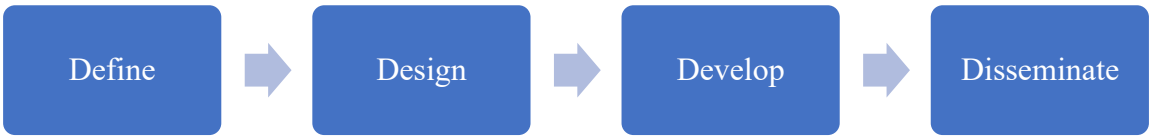


Figure 1. 4-D Model Development Procedure [22]

During the Define stage (three weeks), needs analysis, learner analysis, task and concept analysis, and formulation of learning objectives were carried out. The Design stage (four weeks) involved preparing assessment instruments, selecting media formats, and drafting the initial e-worksheet prototype. The Develop stage (five weeks) included expert validation, product revision, and limited field testing. A one-group pretest-posttest design was implemented in two meetings with 36 eleventh-grade students. The study intentionally concluded at the Develop stage because its primary objective was to produce and validate a feasible instructional product through expert validation and limited classroom testing. Large-scale dissemination, which requires broader implementation across multiple educational contexts, was beyond the scope of this study and is therefore suggested for future research.

The product's validity was assessed by two chemistry education lecturers from Surabaya State University and one high school chemistry teacher. Its practicality and effectiveness were tested with 36 eleventh-grade students from SMAN 1 Gedangan, selected through purposive sampling, during the research period [23]. Before this study, students had not used Liveworksheets or similar interactive digital worksheets, as chemistry instruction relied primarily on PowerPoint presentations, printed worksheets, and conventional methods. Thus, the practicality results reflect students' authentic responses rather than prior familiarity with such platforms. Data in this study were collected using several research instruments, including review sheets to obtain expert suggestions for improving the developed e-worksheet, validation sheets to assess its content and construct validity, student response questionnaires to measure practicality, observation sheets to record students' learning activities, and pretest-posttest instruments

to evaluate students' conceptual understanding and improvements in critical thinking skills after the implementation of the ethnoscience-based e-worksheet.

Validation data were analyzed descriptively using the median of expert ratings on a five-point Likert scale (1 = very bad to 5 = very good) to assess the quality of the developed ethnoscience-based interactive multimedia e-worksheet [24].

Table 1. Validation Score

Score	Criteria
1	Very Poor
2	Poor
3	Moderate
4	Good
5	Very good

The validation data is ordinal in nature. Therefore, it cannot be analyzed using parametric mathematical operations. Consequently, the median value is used as an appropriate measure of central tendency to determine product validity [25]. The e-worksheet is considered valid if the median score obtained for each validation indicator is ≥ 4 [26]. This procedure ensures that the developed learning media meet the required validity standards before being implemented in the classroom. The ethnoscience-based interactive multimedia e-worksheet is considered reliable if the validation instrument demonstrates high internal consistency, indicating stable, consistent results with minimal measurement error [27]. Ensuring instrument reliability strengthens the credibility of validation findings and supports the robustness of conclusions drawn from expert judgment. The following are the levels of reliability [28].

Table 2. Cronbach's Alpha Reliability Criteria

Cronbach's Alpha (α) value	Interpretation
$0.90 \leq \alpha \leq 1.00$	Very high
$0.80 \leq \alpha < 0.90$	High
$0.70 \leq \alpha < 0.80$	Moderate
$0.60 \leq \alpha < 0.70$	Low
$0.50 \leq \alpha < 0.60$	Poor
$\alpha < 0.50$	Very Poor

Pretest and posttest scores were compared to measure the effectiveness of the ethnoscience-based interactive multimedia e-worksheet. This instrument consists of 10 equivalent open-ended questions that assess five critical thinking indicators adapted from Facione: interpretation, analysis, evaluation, explanation, and inference, each represented by two items related to covalent and metallic bonds. Improvement was calculated using the normalized gain (N-gain) score [29].

Before conducting inferential statistical analysis, a normality test was conducted to ensure that the data met the assumptions of parametric testing [30]. Data normality was checked using the Shapiro-Wilk test [31]. If the data were normally distributed ($p > 0.05$), a paired-samples t-test was used to assess whether the improvement in students' critical thinking skills was statistically significant [32]. Conversely, if the data did not meet the assumptions of normality, a nonparametric alternative, namely the Wilcoxon signed-rank test, was applied [33]. This analytical procedure ensures an accurate and reliable evaluation of the effectiveness of the ethnoscience-based interactive multimedia e-worksheet.

$$N - Gain\ Score = \frac{Posttest\ Score - Pretest\ Score}{Ideal\ Score - Pretest\ Score} \qquad [34]$$

To see the categories of magnitude of N-gain score improvement, see Table 3 [35].

Table 3. Criteria for Determining the Level of Effectiveness

N-gain Value	Interpretation
$g \geq 0.7$	High
$0.7 > g \geq 0.3$	Moderate
$g < 0.3$	Low

Furthermore, the data analysis technique used to assess the practicality of the ethnoscience-based interactive multimedia e-worksheet used the Guttman scale. Student responses to the practicality questionnaire were analyzed using a binary scoring system, assigning each "yes" or "agree" response a score of 1 and each "no" or "disagree" response a score of 0 [36]. The total score is then converted into a percentage to determine students' perceptions of the practicality and feasibility of the learning media:

$$Practicality\ percentage = \frac{\sum Respondent\ scores}{\sum Maximum\ Respondent\ scores} x\ 100\% \qquad [37]$$

After being calculated using the formula, the score is then interpreted using Table 4 as follows [35]:

Table 4. Criteria for Determining the Level of Practicality

Percentage (%)	Interpretation
0 – 20	Very Impractical
21 – 40	Not practical
41 – 60	Quite Practical
61 – 80	Practical
81 – 100	Very Practical

3. RESULTS AND DISCUSSION

3.1 Define

The initial stage of this research is the definition stage, which focuses on identifying development needs and analyzing learning needs as the basis for product development [38]. This stage is carried out through a needs analysis that involves literature reviews, analysis of previous research, and pre-research activities, such in interviews with a chemistry teacher at SMA Negeri 1 Gedangan. The interview aims to obtain in-depth information regarding curriculum implementation, learning conditions, media use, and the potential need for innovative learning media in teaching chemical bonds.

Based on the interview results, the RH teacher stated that SMA Negeri 1 Gedangan has implemented the Independent Curriculum. However, students often struggle with chemical bonding due to its abstract nature, which requires them to connect macroscopic phenomena, submicroscopic particle interactions, and symbolic representations. According to cognitive theory in chemistry learning, this representational complexity demands strong prior knowledge, reasoning, and visualization skills, which can hinder the development of critical thinking. Although teachers assess these skills through compound identification and HOTS questions, instruction remains reliant on conventional media such as PowerPoint, whiteboards, Molymod kits, and plasticine, which may reduce engagement. Interviews indicated that an ethnoscience-based e-worksheet, assisted by Liveworksheets, has not yet been implemented, despite teachers recognizing the need for interactive and varied media. Therefore, integrating ethnoscience into multimedia learning is considered a promising approach to promote contextual learning and strengthen critical thinking in the context of chemical bonding. This approach aligns with SDG 4 and the Independent

Curriculum, which emphasizes meaningful, contextual, and student-centered learning to develop students' higher-order thinking skills.

3.2 Design

This stage focused on organizing content and layout in Canva and integrating the ethnoscience approach with critical-thinking indicators into Liveworksheets to produce interactive multimedia. The design process comprised four key steps: constructing a criterion-based test, selecting media, determining format, and developing the initial draft [39]. First, a criterion-based test was developed, based on the learning objectives formulated in the Definition stage, to measure students' critical thinking skills in chemical bonding using pre- and posttest instruments. This test was designed using Facione's critical thinking framework, including interpretation, analysis, inference, evaluation, explanation, and self-regulation, to ensure content validity and accurately assess the effectiveness of the developed e-worksheet in fostering critical thinking skills [40], [41]. Second, media selection identified an ethnoscience-based e-worksheet as the most appropriate learning medium, as it integrates local cultural contexts with scientific concepts. This approach aligns with constructivist learning theory by enabling students to actively construct knowledge through culturally relevant experiences. The e-worksheet operationalizes constructivist principles through structured learning phases that provide scaffolding, problem-based tasks derived from ethnoscience phenomena, and interactive exploration using Liveworksheets features such as drag-and-drop, multiple-choice, and open-response activities, which guide students in gradually developing a meaningful understanding of abstract chemical bonding concepts [7], [42].

Third, the format selection ensures alignment with the chemical bonding material through a storyboard that guides multimedia development. The e-worksheet includes standard components (cover to references) and five ethnoscience learning phases: Review, Task, Solution, Reflection, and Evaluation, which integrate verbal explanations with visual representations. In line with Dual Coding Theory, the design combines textual descriptions, molecular illustrations, and ethnoscience-based cultural contexts to support students in processing information through complementary verbal and visual channels. Cultural phenomena presented through ethnoscience provide concrete visual contexts that help students connect everyday experiences with submicroscopic and symbolic representations of chemical bonding, thereby facilitating a deeper understanding of abstract concepts. The initial design phase produced Draft I of the ethnoscience-based e-worksheet using digital design and interactive worksheet platforms that enabled the integration of visual content and interactive learning activities [43]–[45].



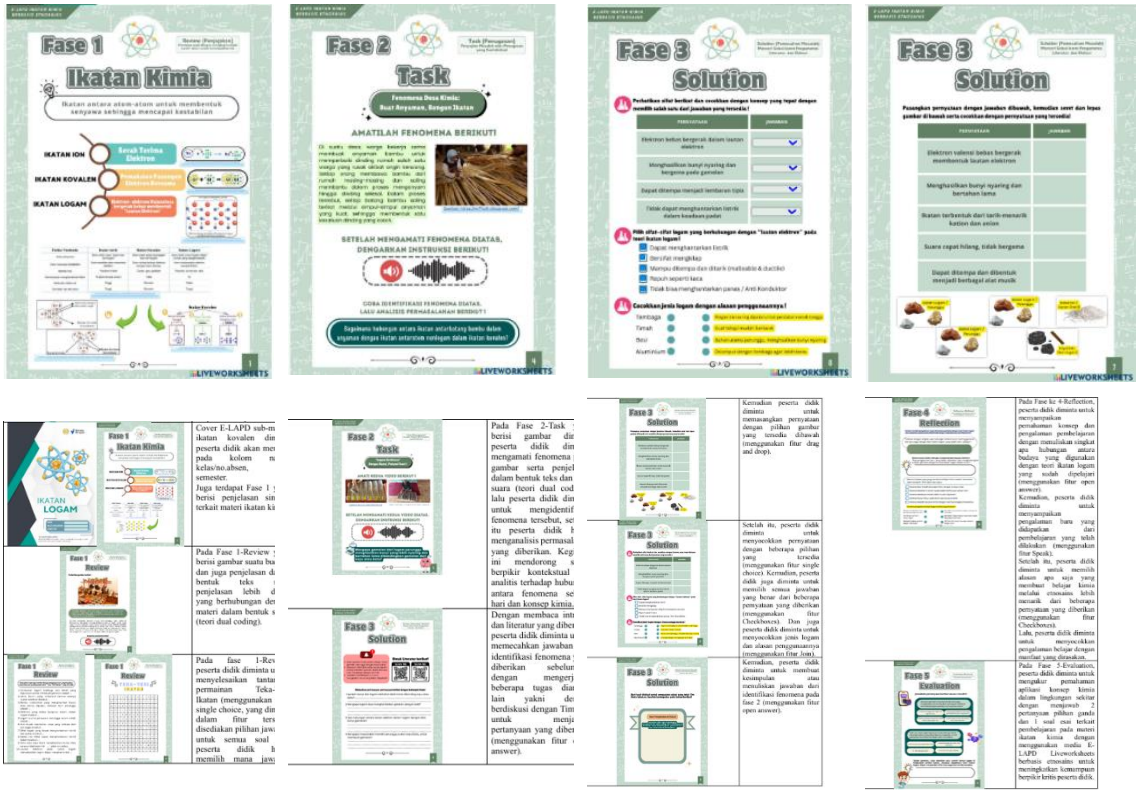


Figure 2. E-worksheet Storyboard

3.3 Development

The development phase transformed the initial design into a functional, interactive, ethnoscience-based e-worksheet. Guided by the storyboard, this stage ensured systematic integration of content, visuals, and interactive elements to support students' critical thinking skills. The layout and learning materials were created using Canva, selected for its flexibility and capacity to produce visually rich digital media [46]. Using Canva, the researchers designed the e-worksheet layout, including the cover, introduction, objectives, concept maps, instructions, ethnoscience-based materials, and activity pages aligned with the learning phases. This design applies Dual Coding principles by integrating verbal explanations with visual representations to support students' understanding of abstract chemical concepts [45], [47].

At this stage, Canva was used to organize abstract chemical bonding concepts into structured and visually engaging materials. Illustrations, diagrams, icons, color schemes, and typography were carefully selected to enhance readability, engagement, and scientific accuracy [48]. The ethnoscience approach integrates local cultural contexts and everyday phenomena related to chemical bonding, enabling students to link scientific concepts with familiar experiences. This contextual visualization reduces cognitive load and deepens conceptual understanding of abstract chemistry topics, consistent with Dual Coding Theory, which emphasizes that coordinated verbal and visual representations enhance cognitive processing and learning [43]–[45]. After completing the visual and content design, the e-worksheet was exported to Liveworksheets to convert static materials into interactive digital worksheets. The platform was selected because it supports interactive learning activities that facilitate active student engagement, formative assessment, and reflective responses, thereby encouraging students to monitor and evaluate their own understanding during the learning process [49]. At this stage, interactive features such as multiple-choice, short-answer, matching, and drag-and-drop tasks were embedded [50].

These activities were designed to align with critical-thinking indicators, including analysis, interpretation, inference, and evaluation, within an ethnoscience context.

Liveworksheets' automated feedback and response recording support constructivist learning by enabling active exploration, reflection, and problem-solving through immediate feedback. Embedding ethnoscience-based tasks in interactive activities encourages students to construct knowledge meaningfully rather than passively receive information, reflecting a student-centered approach [51]–[54]. The development phase produced Draft I of the ethnoscience-based e-worksheet. This functional, interactive worksheet integrates visual design, ethnoscience content, and activities to enhance critical thinking in chemical bonding. The draft was validated by content and media experts to assess its validity and suitability for classroom use. Feedback from this process informed revisions to ensure academic quality and pedagogical soundness before field testing.

3.3.1 E-Worksheet on Chemical Bonding Based on Ethnoscience Assisted by Direct Worksheets

After the e-worksheet product was completed, validation was conducted by two lecturers from Surabaya State University and one chemistry teacher. The validation results are presented in Table 5.

Table 5. Results of Content Validation and Construct Validation

Aspects	Indicator	Median	Category
Contents	The suitability of the media to the criteria related to the material.	5	Very Valid
	The truth of the material concept in THE e-worksheet	5	Very Valid
	Suitability of critical thinking ability indicators	5	Very Valid
Build	Sentence accuracy based on Indonesian spelling conventions (EYD)	4	Valid
	Clarity of writing	5	Very Valid
	Language used	4	Valid
	Media Design	5	Very Valid
	Use of buttons and navigation	5	Very Valid

Based on the validation results presented in Table 5, the content validity of the ethnoscience-based e-worksheet assisted by Liveworksheets shows a median score of 5, which is included in the “Very Valid” category. All content-related indicators, including the suitability of the media to the material criteria, the correctness of the chemical bonding concept, and alignment with the critical thinking skills indicators, obtained a median score of 5. These results indicate that the learning content embedded in the e-worksheet is conceptually accurate, curriculum-relevant, and appropriately designed to facilitate the development of students’ critical thinking skills.

Furthermore, the construct validity results demonstrated strong feasibility in terms of language, clarity, and media design. Indicators related to sentence accuracy based on Indonesian language conventions and language usage achieved a median score of 4 (Valid), while writing clarity, media design, and the use of buttons and navigation features achieved a median score of 5 (Very Valid). Overall, the combined median score for content and construct validity was 5, indicating the e-worksheet is Very Valid. These findings indicate that the developed e-worksheet meets feasibility standards in both content and construct, demonstrating a well-structured, pedagogically appropriate learning design. The high validity also reflects a systematic development process based on the 4D model, indicating that the media are ready for further testing of their practicality and effectiveness [23].

Table 6. Reliability of Media Validation Instruments

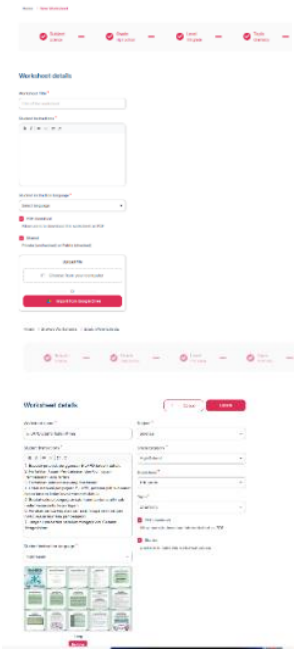
Cronbach's alpha	Number of Items
0.848	8

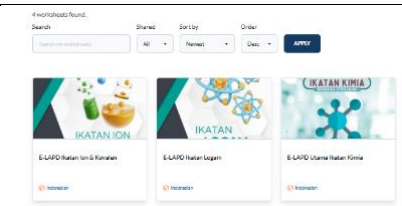
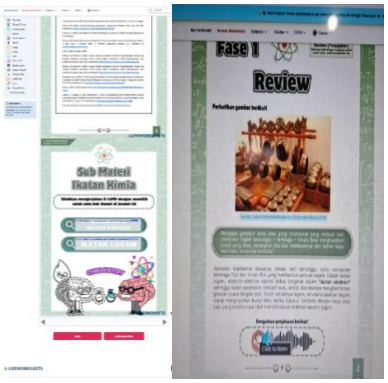
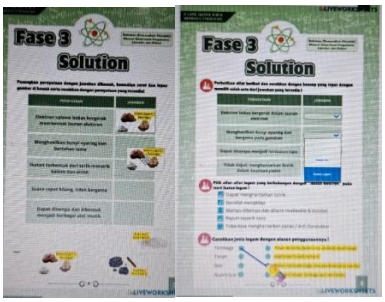
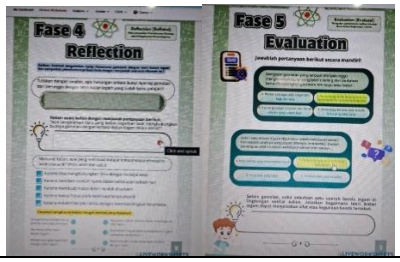
Based on reliability calculations in Jamovi version 2.6.44, a Cronbach's Alpha of 0.848 was obtained, as shown in Table 6. The value of $\alpha = 0.848$ indicates that the media validation instrument has good internal consistency. Based on the interpretation criteria of the Cronbach's Alpha table 2, the reliability value in the range of $0.80 \leq \alpha < 0.90$ is included in the "High" category, which means that each statement item in the instrument has a consistent and mutually supportive relationship in measuring the content and construct of the e-worksheet media.

3.3.2 Product Manufacturing Process Results

The e-worksheet media development process was carried out systematically through several stages, including needs analysis, design, and integration of interactive features. The development utilized Canva for visual design and Liveworksheets to incorporate interactive elements that support student engagement. The resulting product integrates ethnoscience contexts and interactive tasks to facilitate meaningful learning and enhance students' critical thinking skills.

Table 7. E-worksheet Media Development Process

No	Number	Information
1		The initial stage involved analyzing students' learning needs and characteristics to determine the objectives of media development. Next, an e-worksheet framework was prepared, encompassing student identity, learning objectives, usage instructions, ethnoscience-based covalent and metallic bonding materials, student activities, and practice questions designed to foster critical thinking skills. The e-worksheet media was then visually designed using the Canva app, focusing on appearance, readability, and appeal to support student active engagement in the learning process.
2		E-worksheet designs created in Canva are uploaded to the Liveworksheets platform to incorporate interactive features, such as multiple-choice, short-answer, and matching questions, as well as automated feedback, allowing students to interact directly with the medium.

3		A view of three e-worksheet sub-materials designed using Canva after being uploaded to the Liveworksheets platform. At this stage, the learning content and activities are integrated with ethnoscience contexts relevant to students' daily lives to enhance conceptual understanding and critical thinking skills.
4		In terms of interactivity, the e-worksheet features Liveworksheets tools, including short-answer and multiple-choice questions, as well as other interactive features. Furthermore, this ethnoscience-based chemical bonding e-worksheet has added hyperlinks to the sub-topic selection section so students can click them to immediately open a new tab with the sub-topic to be worked on. In addition, this media is enriched with audio features in phase 1, namely in the Liveworksheets play MP4 tool, which adds explanatory voices or instructions to the e-worksheet to make it easier for students to understand the material.
5		The Phase 1 e-worksheet review also included the Liveworksheets "word search" tool, a crossword puzzle titled "Bonding Puzzle." Students selected the correct answer for each question using drop-down menus. These answers were then integrated into a crossword puzzle table, which was then filled in and matched in horizontal and vertical columns according to the instructions. In Phase 2 (Assignment), students observed ethnoscience phenomena related to chemical bonding as an initial learning stimulus. In this phase, the Liveworksheets tool was added, including a YouTube player to display supporting videos, helping students understand the problem's context before moving on to the next phase.
6		In phase 3 (Solution), the e-worksheet includes drag-and-drop, single-choice, checkbox, and merge tools. The drag-and-drop tool is used to match chemical bonding objects or concepts to the appropriate area; the single-choice tool to select the most appropriate answer; the checkbox to select multiple correct answers based on student analysis; and the merge tool to connect pairs of related concepts or statements. The use of these tools aims to train students' conceptual understanding and critical thinking skills interactively.
7		In phase 4 (Reflection), a voice tool allows students to record their reflections on learning in audio. In phase 5 (Evaluation), the e-worksheet includes a selection tool to select the most appropriate answer and an open-ended response tool for free writing to evaluate students' understanding of chemical bonding.

3.3.3 Practical Results of E-worksheet from Ethnoscience-Based Chemical Bonding Assisted by Direct Worksheets

The practicality of the e-worksheet was measured through questionnaire responses from 11th-grade students at SMA Negeri 1 Gedangan. The results of the questionnaire response analysis are presented in Table 8.

Table 8. Practical Results of E-worksheet from Ethnoscience-Based Chemical Bonding

No	Indicator	Percentage (%)	Category
1	Ease of Use of Media	91.1	Very Practical
2	Instructional Uses	99.4	Very Practical
3	Support for Critical Thinking Skills	99.3	Very Practical
	Average	96.6	Very Practical

Based on Table 8, the average practicality score for the ethnoscience-based e-worksheet on chemical bonding was 96.6%, which falls into the "Very Practical" category (based on the P = 81–100 criteria). Each indicator showed a very high percentage, including ease of use of the media (91.1%), instructional usefulness (99.4%), and support for critical thinking skills (99.3%), all of which were categorized as very practical. These results indicate that the developed e-worksheet is easy to use and effective in supporting students' critical thinking skills, as reflected in its high practicality level resulting from the integration of interactive features and ethnoscience contexts that enhance student engagement and learning effectiveness [55]. This supports Dual Coding Theory, which emphasizes that coordinated verbal and visual representations improve learning. However, careful design is essential to maintain clarity and avoid cognitive overload from poorly integrated information [43], [44], [56]–[58].

3.3.4 Results of Ethnoscience-Based E-worksheet Assisted by Liveworksheets in Improving Critical Thinking Skills

A statistical description of students' critical thinking scores in the pretest and posttest following the implementation of the ethnoscience-based e-worksheet is presented in Table 9.

Table 9. Statistical Data Description of Pretest and Posttest Scores

	N	Means	Median	Elementary School	SE
Pretest	36	37.4	37.0	8.62	1.44
Posttest	36	83.4	84.0	6.96	1.16

According to Table 9, the mean pretest score for students' critical thinking skills was 37.4, with a standard deviation of 8.62, indicating that students' initial understanding of chemical bonding was relatively low and varied. The median pretest score of 37.0 further supports that most students began learning with limited prior knowledge. After implementing the ethnoscience-based interactive multimedia e-worksheet, the posttest results showed a substantial increase, with a mean score of 83.4 and a standard deviation of 6.96. The increase in the median score to 84.0 indicates a consistent improvement in student performance after the learning intervention.

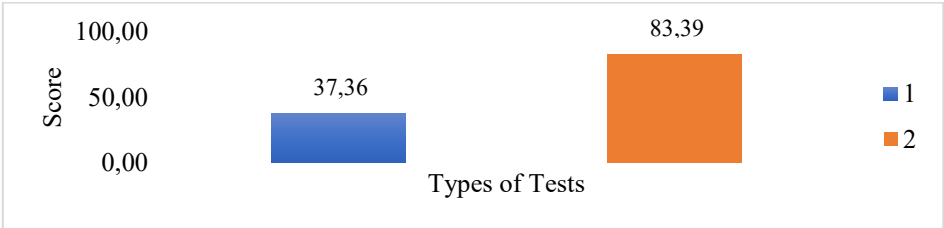


Figure 3. Frequency Distribution Histogram Comparing Pretest and Posttest

Based on Figure 3, the graph compares the average pretest and posttest scores for students' critical thinking skills after implementing the e-worksheet, an interactive multimedia system based on ethnoscience in chemical bond learning. The average pretest score of 37.36 increased substantially to 83.39 in the posttest, indicating a significant increase in students' critical thinking skills after using the e-worksheet interactive media.

Table 10. Results of the Normality Test for Pretest and Posttest Scores

			W	P
Pretest	-	Posttest	0.968	0.362

Note: A low p-value indicates a violation of the normality assumption

Based on the Shapiro–Wilk normality test, the posttest data yielded a significance value of 0.362 ($p > 0.05$), indicating that the posttest scores were normally distributed. This result indicates that the normality assumption is met. Therefore, the pretest and posttest data meet the requirements for further analysis using parametric statistical tests. Consequently, to examine the difference between students' pretest and posttest scores after the implementation of the e-worksheet, an interactive multimedia-based on ethnoscience, a paired-samples t-test was used.

Table 11. Paired Sample T-Test

Comparison	t	df	p	Mean Difference	SE	95% CI (Lower–Upper)	Effect Size (Cohen's d)
Pretest–Posttest	-30.9	35	<0.001	-46.0	1.49	-49.1--43.0	-5.14

Note: H_a: $\mu_1 - \mu_2 \neq 0$

Based on the results of the paired sample t-test, there was a statistically significant difference between students' pretest and posttest scores after the implementation of the ethnoscience-based e-worksheet interactive multimedia. The analysis yielded a t-value of −30.9 with 35.0 degrees of freedom and a p-value <0.001, indicating a significant improvement in student performance. The mean difference between pretest and posttest scores was −46.0, with a 95% confidence interval of −49.1 to −43.0, confirming substantial improvement in learning outcomes. In addition, the effect size (Cohen's d = −5.14) indicated a very strong effect of the learning intervention.

The effectiveness of the ethnoscience-based interactive e-worksheet was evaluated by comparing pretest and posttest scores on students' critical thinking in covalent and metallic bonding. Improvement was measured using the N-gain score to determine the media's effectiveness in enhancing critical thinking [59]. The following is a graph of the improvement categories on the N-gain in Figure 4.

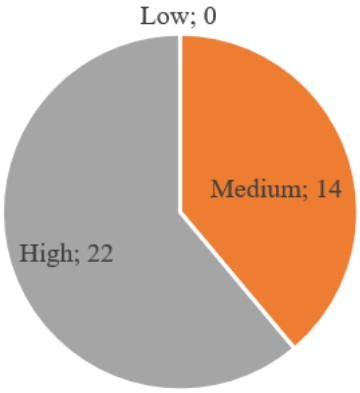


Figure 4. N-gain Score Graph

N-gain measures effectiveness based on pretest–posttest differences. No students were in the low category, indicating overall improvement in critical thinking skills. About 39% were in the medium category, while 61% achieved high N-gain scores. These results suggest that most students experienced significant gains in critical thinking through the interactive e-worksheet.

Table 12. Results of N-gain to Determine the Level of Effectiveness

N-gain Value	Interpretation
0.73	High

Based on effectiveness criteria ($g \geq 0.70$), this value falls into the high category, indicating that the ethnoscience-based interactive e-worksheet was highly effective in improving students' critical thinking on chemical bonding, particularly covalent and metallic bonds. This result is consistent with the N-gain distribution, in which most students were in the high category and none in the low category, indicating consistent learning improvement. These findings are consistent with previous studies showing that ethnoscience-based learning and interactive multimedia can significantly enhance students' conceptual understanding and higher-order thinking skills in chemistry learning.

These findings align with contextual and constructivist learning theories, which emphasize that learning is more effective when learning materials are connected to real-life and cultural contexts and when learners actively construct their knowledge. The integration of ethnoscience elements and interactive features in e-worksheet promotes student-centered learning, increases motivation and cognitive engagement, and facilitates meaningful learning experiences [7], [42], [53], [60]. Therefore, the use of ethnoscience-based interactive e-worksheet multimedia is considered effective in fostering meaningful learning and enhancing students' critical thinking skills [37].

This research demonstrates that the Liveworksheets-assisted, ethnoscience-based e-worksheet is highly effective in enhancing students' critical thinking skills in chemical bonding, particularly in covalent and metallic bonds, as indicated by significant improvements between pretest and posttest scores. The interactive tasks and immediate feedback provided through the platform encourage students to actively engage in interpretation, analysis, evaluation, inference, and explanation, which are key components of critical thinking according to Facione's framework [40], [61], [62]. These activities enable students to connect ethnoscience contexts to abstract chemical concepts, transforming complex representations of bonding into meaningful learning experiences that promote higher-order thinking.

These results are consistent with previous studies reporting that ethnoscience-based learning can enhance students' engagement and critical thinking in science education [11], [12], [63]. At the same time, interactive multimedia supports deeper conceptual understanding of abstract chemistry topics through visual and interactive representations [18], [44]. By integrating cultural contextualization with digital interactivity, the developed e-worksheet provides a learning environment that not only facilitates conceptual understanding of chemical bonding but also systematically strengthens students' critical thinking skills in chemistry learning.

3.3.5 Results of Ethnoscience-Based E-worksheet Assisted by Liveworksheets in Improving Critical Thinking Skills in Each Indicator

This section presents an N-gain analysis for each critical thinking indicator, including interpretation, analysis, evaluation, explanation, and inference, after implementing ethnoscience-based e-worksheet assisted by Liveworksheets in chemical

bonding learning. This analysis compares pretest and posttest scores to determine improvement and highlight the media's contribution to specific aspects of critical thinking.

Table 13. The N-gain Results of Each Indicator to Determine the Level of Effectiveness

Indicator	N-gain Value	Interpretation
Interpretation	0.77	High
Analysis	0.60	Medium
Evaluation	0.61	Medium
Explanation	0.85	High
Inference	0.83	High

Based on the calculation results, Table 13 presents the N-gain results for each critical thinking indicator after the implementation of the ethnoscience-based e-worksheet assisted by Liveworksheets. The interpretation indicator obtained an average N-gain of 0.77, calculated from the increase between the pretest (42.71) and posttest (87.15) scores, which falls into the high category ($g \geq 0.70$). Similarly, the explanation and inference indicators achieved high N-gain scores of 0.85 and 0.83, respectively. These results indicate that the interactive e-worksheet was highly effective in enhancing students' abilities to interpret chemical bonding concepts, articulate logical explanations, and draw appropriate conclusions.

In contrast, the analysis (0.60) and evaluation (0.61) indicators were classified in the moderate category ($0.30 \leq g < 0.70$). Although the gains have not yet reached the high category, the results still demonstrate meaningful improvement in students' abilities to analyze relationships among chemical bonds and evaluate problem-solving processes. The ethnoscience-based e-worksheet particularly strengthened interpretation, explanation, and inference skills, while analysis and evaluation skills showed moderate but promising development. These findings are consistent with previous studies indicating that ethnoscience-based learning can enhance students' critical thinking and conceptual understanding in science education [11], [12], [63], and that interactive multimedia supports deeper reasoning and engagement in learning abstract chemistry concepts [18], [44].

The improvement in inference skills also aligns with the N-gain results, indicating that integrating cultural contexts and interactive tasks can effectively support the development of critical thinking in chemistry learning. However, the use of a one-group pretest–posttest design and a limited sample size restricts the generalizability of the findings. Therefore, future studies are recommended to involve larger samples, include control groups, and examine the effectiveness of this approach across different chemistry topics.

4. CONCLUSION

This study shows that an ethnoscience-based e-worksheet on chemical bonding, supported by Liveworksheets, is a valid, practical, and effective learning medium for improving high school students' critical thinking skills in chemical bonding. The integration of ethnoscience context with interactive digital features helps students engage meaningfully with abstract concepts, particularly covalent and metallic bonding. These findings suggest that ethnoscience-based learning multimedia can serve as an effective teaching strategy to foster higher-order thinking in chemistry learning.

These findings contribute to chemistry education by providing empirical support for integrating ethnoscience and digital interactive media within a coherent instructional design framework. Unlike many previous studies that examine ethnoscience or digital

media separately, this study provides indicator-based evidence of improved critical thinking, supported by strong statistical results. The findings show that contextual tasks and immediate feedback within the e-worksheet can systematically strengthen multiple indicators of critical thinking. Moreover, the developed e-worksheet aligns with the principles of the Merdeka Curriculum by promoting contextual, student-centered, and inquiry-oriented learning, while also supporting the SDG 4 agenda of improving the quality and relevance of education through innovative learning media. Thus, the developed e-worksheet represents not only a practical instructional innovation but also a theoretical contribution to culturally responsive and technology-enhanced approaches for fostering critical thinking in chemistry learning.

In terms of prospects, this learning medium shows strong potential for broader application in chemistry education. Future research may explore its use across different chemistry topics, educational levels, and learning contexts, and examine its long-term impact on students' critical thinking and other 21st-century skills. These directions may further support the development of culturally responsive and interactive digital learning media in science education.

AUTHOR CONTRIBUTION STATEMENT

ZS conceptualized the research, designed and developed the ethnoscience-based e-worksheet using Liveworksheets, collected and analyzed the data, and drafted the manuscript. SS supervised the research process, contributed to the research design and methodology, validated the learning media and instruments, reviewed and revised the manuscript, and approved the final version for publication.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest in the conduct and publication of this research.

DECLARATION OF USE OF AI IN SCIENTIFIC WRITING

The authors used ChatGPT (OpenAI) only to assist in checking grammar and improving English language expressions. All ideas, analyses, data interpretations, and conclusions are entirely the authors' own work, and the authors take full responsibility for the content of this article.

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